

NEW STUYAHOK, AK, USA

WMO: 702697

Lat: 59.450N

Lon: 157.333W

Elev: 92

StdP: 100.22

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00107

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.3	-24.0	-30.3	0.2	-25.8	-28.4	0.3	-23.3	16.4	0.9	15.2	1.8	5.0	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	23.3	15.2	21.0	14.1	18.9	13.2	16.0	21.7	14.9	19.4	14.0	17.7	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.0	10.1	17.3	13.0	9.4	15.5	12.3	9.0	14.7	44.9	22.0	41.7	19.4	39.3	17.8	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-8.3	-7.9	-6.4	-0.5	6.5	10.7	13.0	12.7	8.6	1.9	-7.0	-9.1
	DBStd	10.12	8.18	7.78	6.78	4.83	3.30	2.97	2.87	2.22	3.22	4.89	6.72	8.22
	HDD10.0	3454	569	501	509	314	118	22	5	3	61	250	511	591
	HDD18.3	6244	827	734	768	564	368	229	169	174	292	508	761	849
	CDD10.0	256	0	0	0	0	9	44	96	88	19	0	0	0
	CDD18.3	4	0	0	0	0	0	2	3	0	0	0	0	0
	CDH23.3	71	0	0	0	0	2	28	38	2	0	0	0	0
	CDH26.7	12	0	0	0	0	0	7	6	0	0	0	0	0
Wind	WSAvg	4.5	5.6	5.0	4.8	4.5	4.4	3.7	3.7	3.8	4.0	4.6	4.9	5.1
Precipitation	PrecAvg	826	62	53	47	44	48	55	77	83	108	89	79	80
	PrecMax	1043	152	97	89	69	100	83	122	142	180	156	184	148
	PrecMin	625	19	12	11	6	17	18	39	27	45	13	10	36
	PrecStd	112	28	19	25	15	20	17	21	27	33	31	41	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	4.9	7.6	12.2	22.0	28.1	27.1	22.7	19.9	12.0	7.1	4.6
		MCWB	4.4	3.1	4.2	6.3	12.0	16.4	18.4	15.8	13.4	9.0	6.0	3.3
	2%	DB	4.3	3.5	4.9	9.2	18.0	23.0	23.9	20.8	16.4	10.5	5.2	3.4
		MCWB	3.3	2.1	2.9	4.7	10.5	14.0	16.0	15.0	11.7	8.1	4.6	2.2
	5%	DB	3.2	2.5	3.1	7.3	15.1	20.1	21.2	18.9	14.6	9.3	3.6	2.6
		MCWB	2.4	1.2	1.4	3.8	8.8	12.9	14.7	13.7	11.2	7.5	2.9	1.6
	10%	DB	2.3	1.4	2.0	5.5	12.5	17.4	18.9	17.5	13.0	8.3	2.1	1.8
		MCWB	1.6	0.4	0.6	2.8	7.4	11.4	13.8	13.0	10.6	7.0	1.6	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	3.3	4.5	6.7	13.7	16.7	19.2	17.1	14.0	9.9	6.5	3.7
		MCDB	6.4	4.4	6.7	11.1	20.8	26.7	26.6	20.5	17.1	10.7	7.2	4.2
	2%	WB	3.5	2.3	3.1	5.1	10.8	14.7	16.7	15.4	12.8	8.7	4.7	2.6
		MCDB	4.3	3.3	4.6	8.6	17.4	22.2	22.4	18.8	15.2	9.6	5.1	3.2
	5%	WB	2.5	1.3	1.9	4.0	9.1	13.2	15.3	14.6	11.9	7.9	3.0	1.7
		MCDB	3.2	2.2	3.1	6.7	13.9	19.3	20.2	17.7	13.7	8.9	3.5	2.4
	10%	WB	1.5	0.5	0.7	3.0	7.9	11.9	14.2	13.8	11.1	7.2	1.5	0.9
		MCDB	2.2	1.3	1.6	5.2	11.8	16.5	18.0	16.5	12.7	8.2	2.0	1.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	7.4	7.4	8.9	9.3	7.9	7.4	7.2	6.0	6.3	5.7
		MCDBR	4.8	5.2	6.7	8.9	13.9	14.6	13.1	10.7	9.1	5.9	6.4	5.7
		MCWBR	4.2	4.5	5.0	5.6	7.5	7.0	6.3	5.5	5.4	4.4	5.9	5.2
	5% WB	MCDBR	4.8	5.2	6.3	8.0	12.1	13.5	11.8	8.7	7.4	5.1	6.0	5.4
		MCWBR	4.3	4.8	4.9	5.2	6.8	6.8	6.1	4.9	4.9	4.1	5.7	5.0
Clear-Sky Solar Irradiance	taub	0.252	0.297	0.307	0.342	0.377	0.362	0.362	0.356	0.329	0.302	0.306	0.247	
	taud	2.118	2.168	2.196	2.238	2.283	2.401	2.439	2.454	2.500	2.424	2.018	1.975	
	Ebn at Noon	540	701	823	857	849	871	862	840	808	712	419	367	
	Edh at Noon	45	75	99	114	118	107	101	92	75	59	51	35	
All-Sky Solar Radiation	RadAvg	0.41	1.08	2.81	4.26	5.01	5.33	4.37	3.60	2.54	1.32	0.54	0.24	
	RadStd	0.06	0.15	0.38	0.44	0.56	0.53	0.48	0.43	0.34	0.14	0.08	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	+2.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page